

Inference Distribution Network Requirements and Architecture

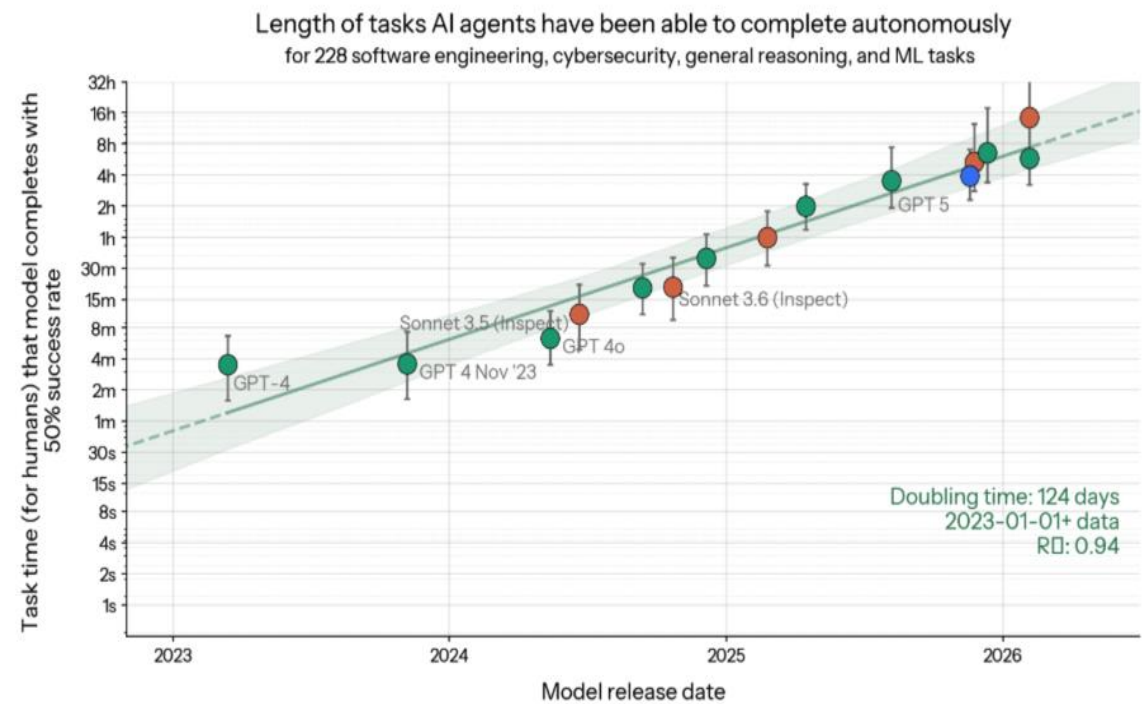
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China Mobile

AI Trends: Evolving Exponentially

METR

The length of tasks AI can complete autonomously doubles **every 7 months** over past 6 years

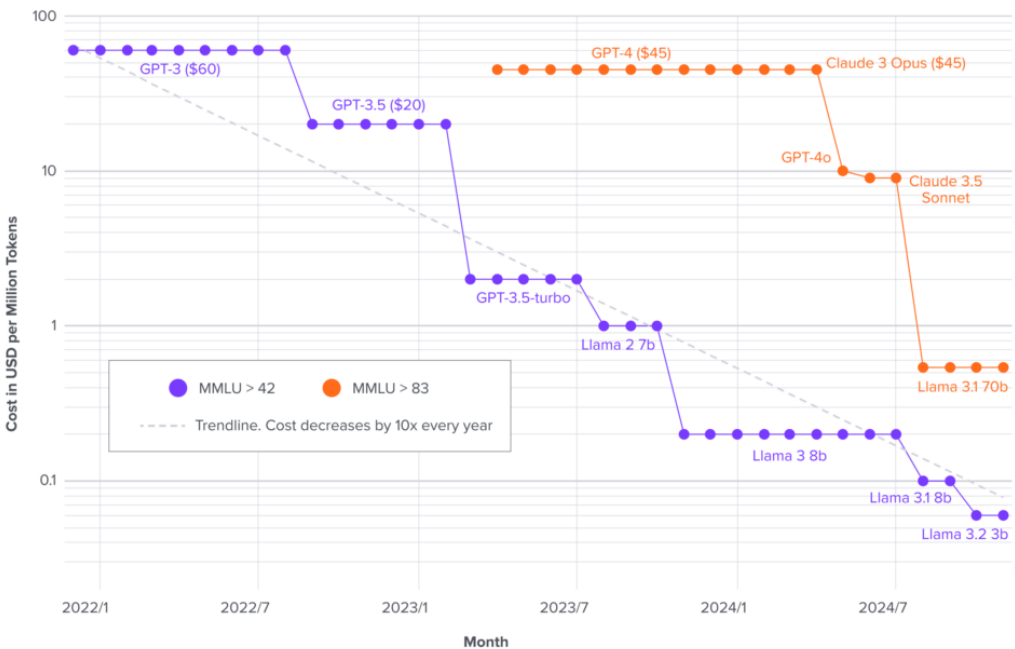


The length of tasks (measured by how long they take human professionals) that generalist frontier model agents can complete autonomously with 50% reliability has been doubling approximately every 7 months for the last 6 years. The shaded region represents 95% CI calculated by hierarchical bootstrap over task families, tasks, and task attempts.

A16Z

Inference API costs decrease by **10x every year**

Cost of the Cheapest LLM with a Minimum MMLU Score (Log Scale)



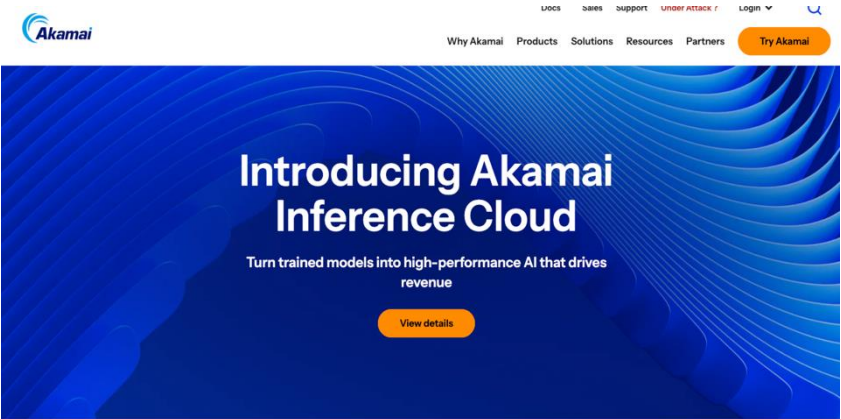
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More From this Contributor

Industry View: Distributed Network Architecture



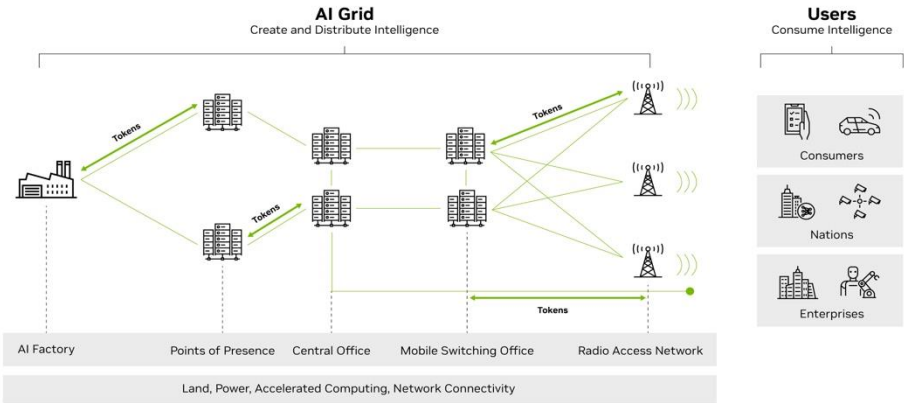
Akamai:

Distributed Inference using 4,200+ global CDN edge nodes



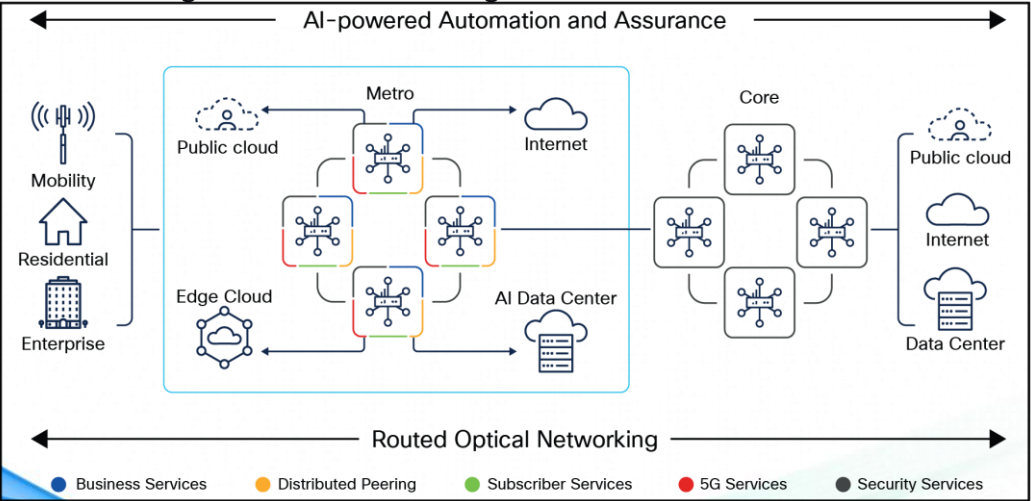
NVIDIA: AI Grid

Distributing intelligence across widespread areas via decentralized computing



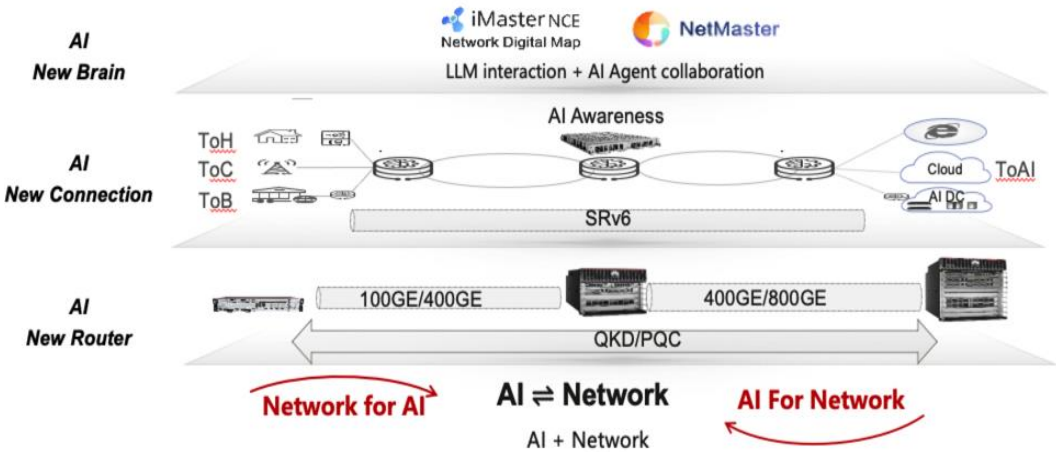
Cisco:

Agile Services Networking Solution for Carrier network



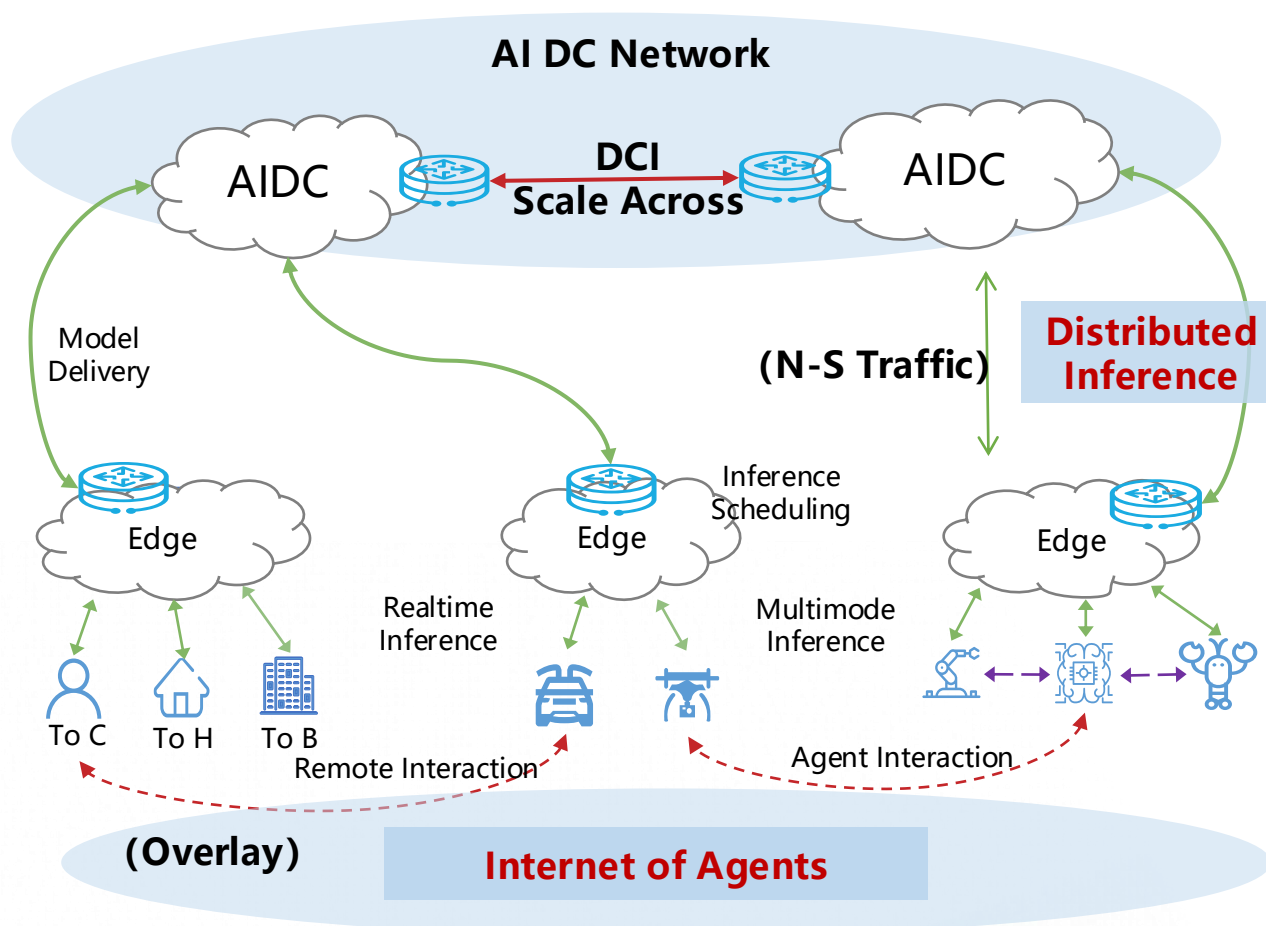
Huawei:

AI WAN solution for intelligent transformation



Intelligence is moving to the edge, and the network provider should take more roles to improve AI experience

AI is restructuring network traffic



North-South Traffic Growth:

- Collaborative inference
- Model delivery
- Real time inference

East-West Traffic Growth:

- Scale-Across
- User-Agent-Tools interaction
- Remote interaction
- Agents' communication

High Performance Requirements:

- Billions level concurrence
- Realtime **stable** low latency
- High frequency interaction in IOA
- Multimode complex inference
- Upstream bandwidth

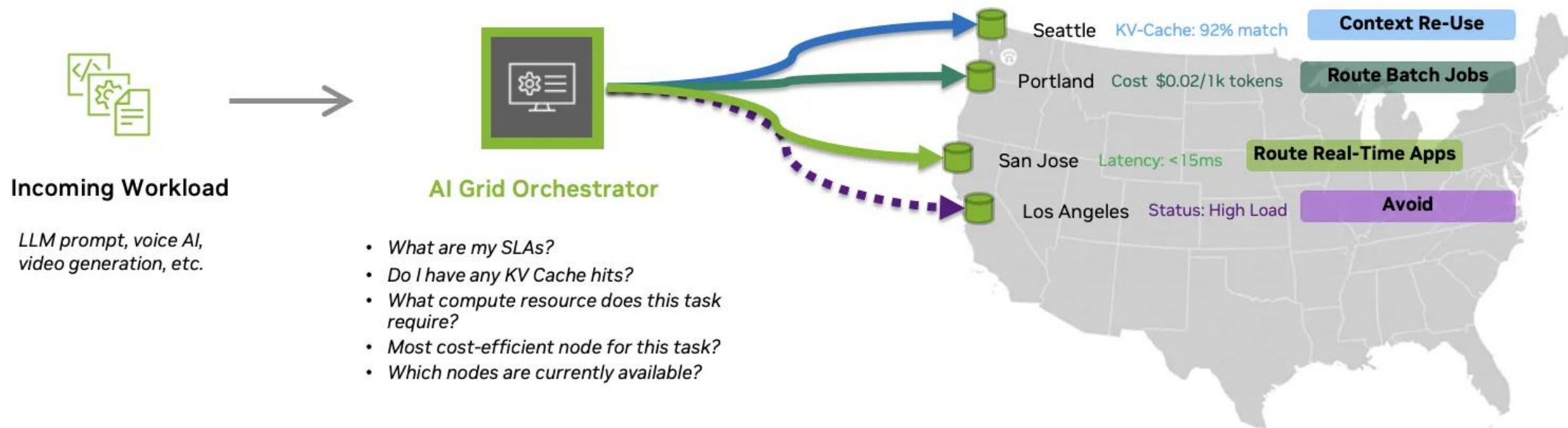
AI and agentic AI are projected to increase consumer-driven network traffic by about **6.6x**, making this the dominant driver of overall internet traffic expansion. AI inference traffic is projected to be **25% of total network traffic by 2035**.

----Cisco Report <AI Impact on Wide Area Networks>

Use case: AI job and resource aware routing

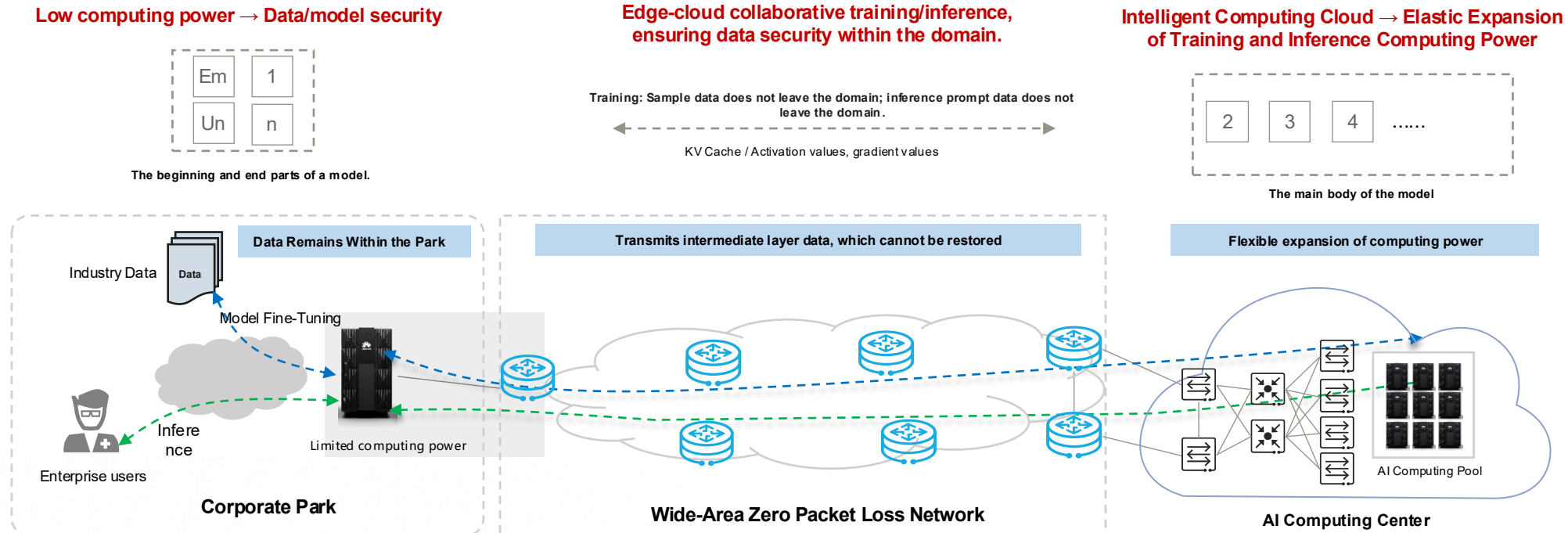
AI Grid Control Plane

Workload-aware, resource-aware, and KPI-aware orchestration



- No standard interface between AI orchestration and network control plane
- Application-network collaborative awareness is missing, both control and forwarding planes need APIs for workload/resource visibility

Use case: Collaborative Distributed Inference



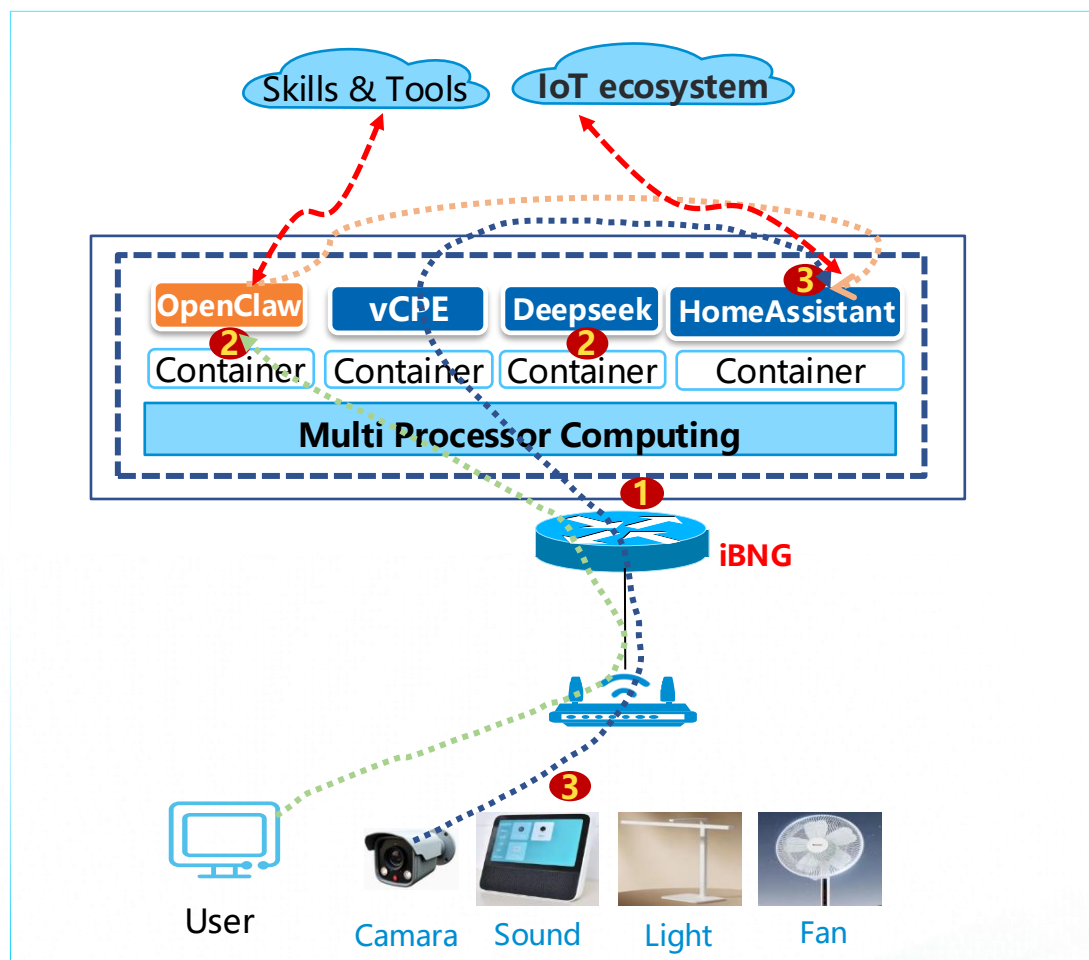
Enterprise side: Deploy a small amount of computing power. Use the PD separation + Split Learning and intelligent computing center to perform edge-cloud collaborative inference, ensuring that Prompt does not leave the campus.

WAN: Intelligent computing private lines are deployed to connect enterprise computing power and intelligent computing center servers, ensuring lossless and high-throughput transmission over the WAN of hidden variables.

Intelligent computing center: On-demand intelligent computing servers are leased. Prefill & Decode instances are deployed during service inference, and hidden variables are received and sent.

- Lossless WAN transport is not guaranteed (1e-7 required vs. 1e-5 today)
- Cost-effective lossless solutions are also needed

Use case: Smart Home with intelligent BNG



01

iBNG with MPC

- iBNG (intelligent BNG) with built-in MPC (Multi-Processor Computing) board provides computing/storage power inside the router

02

Deploy claw and LLM in Container

- Containerized OpenClaw, vCPE, and DeepSeek models on MPC, connected to open source Home Assistant

03

Control the smart IoT via claw and home assistant

- User can control smart iot elements with one portal
- More smart home scenarios can be developed and added as .skill in openclaw
- Less latency, more secure, more easy deployment

- MPC computation power, MPC resource isolation, container interfaces, and security assurance on iBNG require careful design

Gap 1 – Orchestration & Network Interface

- No open interface between AI orchestration and network
- Missing collaborative awareness (control & data plane)
- Prevents workload-aware routing across multi-vendor networks

Gap 2 – Lossless WAN

- Lossless transport at scale is not available
- Today's carrier networks cannot deliver stable $1e-7$ loss rate
- Cost-effective solutions are also needed

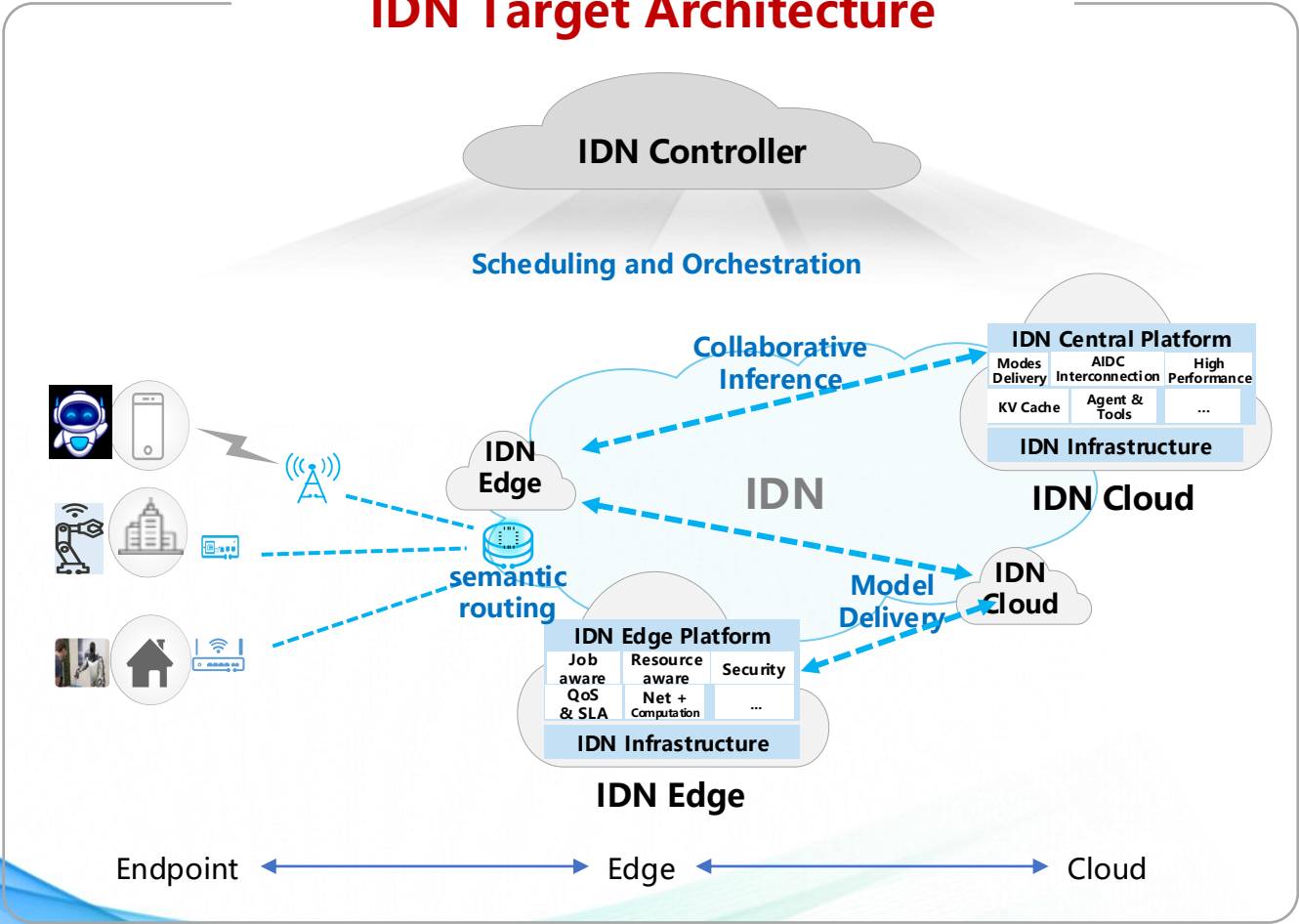
Gap 3 – Edge Inference on iBNG

- Resource isolation
- Container management
- Security assurance

Inference Distribution Network: shifting towards inference

A distributed collaborative architecture with edge intelligence, dynamic scheduling, and inference service recognition to provide high concurrency, low latency, high performance network guarantees.

IDN Target Architecture



IDN Services

Intelligent element distribution and supply



For intelligent elements such as models and agents, unified management, scheduling, distribution, and deployment optimization create a high-performance Token supply platform.

Unified access and semantic routing



Through distributed intelligent gateway (iBNG) access and multi-dimensional SLA-based routing decisions, IDN Services deliver cost-effective, latency-optimal, and stable Token consumption services for intelligent inference applications.

Summary

- Inference traffic is becoming the dominant workload on the Internet
- New AI services are reshaping how we think about network architecture, traffic patterns, and service models
- *<DIN Problem Statement, Use Cases, and Requirements>* (draft-song-rtgwg-din-usecases-requirements-02)
- IDN Architecture Framework: we are working on a framework that covers the service, scheduling, and infrastructure layers for intelligence delivery

Challenges

- System-level efficiency matters, not only per-task SLA
- Lossless transport over WAN is not optional
- Collaboration between networking and inference is essential
- Edge real-time inference demands deterministic latency and jitter control

Standardization Gaps

- How network aware inference job requirements
- Protocols between AI-orchestration control planes and network forwarding
- Cost-effective lossless WAN
- iBNG interfaces for resource isolation, container management, security, etc.
- Need to deeply analyze AI inference new use case requirements and the technical gap